1. Problem Statement: Why Does This Matter?

Good morning everyone.

Our research focuses on human blastocyst classification using deep learning.

In IVF, selecting an embryo with the potential to develop successfully is an important part of the treatment process. Embryologists examine embryo morphology to assess its quality, but this process requires expertise and can involve subjective variation between observers.

Our goal is to develop an AI-based system that can assist embryologists by classifying blastocyst images into seven morphological categories: AA, AB, BA, BB, BC, CA, and CB.

The seriousness of this problem lies in its potential impact on reproductive treatment. However, our model is designed as a research and decision-support tool, not as a replacement for embryologists or a guarantee of implantation success.

2. Why Did We Choose This Solution?

We developed a hybrid deep learning architecture combining VGG16 and Swin Transformer.

VGG16 learns local visual features such as edges, textures, and cellular structures. Swin Transformer captures broader spatial relationships and hierarchical visual patterns.

By combining their feature representations, we aim to capture complementary information that a single architecture might miss.

Our hypothesis is that combining convolutional features with transformer-based features could improve morphological classification. Our current results are promising enough to justify further investigation, but they do not yet prove that the hybrid model is consistently superior.

3. Why Is the Accuracy Limited?

Our current hybrid model achieved approximately 72.73% test accuracy on a test set of 22 images. Earlier experiments produced results around 86% on a different test split.

The main challenge is our limited dataset. We have approximately 142 images, and the number of examples differs substantially between classes. For example, AA has many more images than rare categories such as CA and CB.

Deep learning models usually benefit from larger and more diverse datasets. With limited examples, the model may learn patterns specific to the training images rather than general features that transfer reliably to unseen embryos.

Our test set is also very small, so each individual prediction significantly affects the accuracy. Therefore, we must interpret the current results cautiously.

Importantly, our evaluation includes macro precision, macro recall, macro F1, ROC-AUC, and a confusion matrix. These metrics help us investigate performance beyond accuracy, particularly for imbalanced classes.

4. Data Cleaning and Preprocessing

Before training, we organized the images according to their seven target classes and checked the dataset structure and labels.

We then split the data into training, validation, and test sets. The training set was used to learn model parameters, the validation set to select checkpoints and monitor performance, and the test set to evaluate the selected model.

We applied image transformations during preprocessing. Training transformations can introduce moderate variations that help reduce overfitting, while validation and test images receive consistent evaluation preprocessing.

The images were converted into tensors and processed in batches. We also used pretrained models, class-weighted cross-entropy loss to address class imbalance, and a two-stage training strategy.

In the first stage, we trained the projection layers and classification head while freezing the pretrained feature extractors. In the second stage, we fine-tuned the model using lower learning rates for the pretrained backbones.

These techniques help make better use of limited training data, although they cannot completely overcome data scarcity.

5. How Can We Improve the Model?

Our next steps are to collect more independently sourced embryo images, particularly for underrepresented classes, and improve the reliability of our train-validation-test splits.

We can investigate carefully selected data augmentation, compare VGG16, Swin Transformer, and the hybrid architecture under identical experimental conditions, and perform grouped cross-validation where images from the same embryo remain in the same fold.

We should also examine misclassified images, improve label consistency with expert review, and report confidence intervals and per-class performance.

Finally, external validation on images from different clinics would help determine whether the model generalizes beyond our current dataset.

6. Why Should This Become a Research Paper?

This project addresses a meaningful problem at the intersection of artificial intelligence and reproductive medicine.

Its research contribution is not simply achieving a high accuracy. It is investigating whether combining convolutional neural networks and transformer-based visual features can support blastocyst morphology classification when data is limited and classes are imbalanced.

A research paper can document the architecture, preprocessing, experimental protocol, evaluation metrics, limitations, and comparison with baseline models. It can also provide a reproducible starting point for other researchers.

Further studies should examine larger datasets, expert-validated labels, external clinical validation, model explainability, and—only with suitable outcome data—whether predictions are associated with implantation or live-birth outcomes.

7. Conclusion

To conclude, this is an early-stage research prototype with meaningful potential, but it is not yet a clinically validated system.

Our present results establish a starting point for systematic investigation. With larger datasets, rigorous evaluation, and collaboration with embryologists, this work could contribute to the development of more consistent and evidence-based embryo assessment tools.

Thank you.